

A two-coordinate obstruction for affine interpretations of the Yolcu–Aaronson–Heule Collatz system

Lucas Valbuena

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Abstract

We prove a restriction on matrix-interpretation proofs for the eleven-rule Collatz system constructed by Yolcu, Aaronson, and Heule. For nonnegative real affine maps in two coordinates, assume that every symbol matrix has first diagonal entry at least one. If all eleven rules are weakly oriented coefficientwise, every first-coordinate offset gap is zero. Thus no rule can be removed in a first step using this strictly monotone interpretation format, regardless of coefficient size. The same conclusion holds after word reversal and allows arbitrary admissible boundary maps. The proof includes singular matrices and zero secondary entries. Lean 4 checks the complete theorem and its supporting algebra. The relaxed TOP setting and arctic interpretations are outside the theorem’s scope.

1 Introduction

Yolcu, Aaronson, and Heule constructed a mixed binary–ternary string rewriting system whose termination is equivalent to the Collatz conjecture [1, Theorem 3.17]. Their construction turns a question about integer trajectories into a termination problem for eleven rules. The system, its arithmetic interpretation, and that equivalence are their results. This paper studies a restriction on possible matrix-interpretation proofs for their system.

Rule removal builds a termination proof in stages. An interpretation orients all remaining rules weakly and a nonempty subset strictly; relative termination then permits removal of that subset. Repeating the process can establish termination [1, Theorems 2.6 and 2.15]. The first step therefore needs at least one strict rule while retaining weak comparisons for the complete system.

Our result rules out that first step in two coordinates for the strictly monotone affine format defined below. It quantifies over all admissible nonnegative real coefficients, including irrational ones, and imposes no upper bound. A failed bounded SAT search cannot establish this conclusion: it leaves larger coefficients and coefficients outside its encoding unexamined. The obstruction is algebraic. It forces all eleven candidate strict gaps to vanish, rather than excluding one chosen rule or one coefficient range.

Dimension restrictions matter in this setting. Yolcu, Aaronson, and Heule report that, for some weakened variants, their tool found only interpretations of dimension five [1, p. 4]. Their Table 1 also records a natural dimension-five certificate for the subsystem omitting $bf \rightarrow ga$ in our notation. These are search outcomes, not proofs of minimal dimension. The present result supplies a coefficient-independent exclusion for the full system in dimension two, within a narrower, precisely stated interpretation format.

The proof has two parts. First, the matrix comparisons force the five digit matrices to share a triangular orientation, with unit first diagonals. Second, an affine analysis forces every first-offset gap to zero. The upper triangular case requires retaining products between matrix entries and offsets, including several zero-slope cases. All new obstruction results are formalized in Lean. Appendix A reproduces the definitions and the exact final theorem type.

2 The system and the interpretation format

We use the alphabet $\{a, b, c, d, e, f, g\}$ and the ASCII names from the authors' implementation [2]. Table 1 relates these names to the symbols in their article. The rules are their full mixed-base system [1, Section 3.2], with the leftmost symbol interpreted as the outermost map:

$$\begin{aligned}
 ad &\rightarrow d, & bd &\rightarrow gd, \\
 ae &\rightarrow ea, & af &\rightarrow eb, & ag &\rightarrow fa, \\
 be &\rightarrow fb, & bf &\rightarrow ga, & bg &\rightarrow gb, \\
 ce &\rightarrow cb, & cf &\rightarrow caa, & cg &\rightarrow cab.
 \end{aligned} \tag{1}$$

This paper and the ASCII implementation	a	b	c	d	e	f	g
Yolcu–Aaronson–Heule [1]	f	t	◁	▷	0	1	2

Table 1: Symbol correspondence. The letters a, b are binary digits, e, f, g are ternary digits, and c, d mark the boundaries.

The five symbols a, b, e, f, g are called digits; c, d are boundary symbols. Assign to each symbol s an affine map $F_s(x) = M_s x + v_s$ on $\mathbb{R}^2$. An interpretation is *admissible* when

$$M_s \in \mathbb{R}_{\geq 0}^{2 \times 2}, \quad v_s \in \mathbb{R}_{\geq 0}^2, \quad (M_s)_{00} \geq 1 \quad \text{for every } s. \tag{2}$$

For words, composition determines M_w, v_w ; in particular, $M_{uv} = M_u M_v$ and $v_{uv} = M_u v_v + v_u$. A rule $\ell \rightarrow r$ is *weak* if $M_\ell \geq M_r$ and $v_\ell \geq v_r$ entrywise. Its first-offset gap is $\text{gap}(\ell, r) = (v_\ell)_0 - (v_r)_0$. The strictness criterion considered here is $\text{gap}(\ell, r) > 0$.

Theorem 1. *For every admissible interpretation in two real coordinates, weak orientation of all eleven rules in (1) implies $\text{gap}(\ell, r) = 0$ for every rule. The same assertion holds for the eleven rules $\text{rev}(\ell) \rightarrow \text{rev}(r)$, where each word is reversed and the direction of every rule is retained.*

The two orientations are separate implications, not simultaneous hypotheses. There is no coefficient bound, integrality requirement, nonsingularity assumption, or prescribed boundary shape. The conclusion concerns the first offsets; it does not assert equality of the complete affine maps. The following sections describe the checked proof. The formal artifact contains the full algebraic case analyses behind the compressed arguments below.

2.1 Scope and soundness

The theorem covers the strictly monotone, full-context interpretation format specified in (2) only. Every symbol, including both boundaries, must satisfy $(M_s)_{00} \geq 1$. The relaxed TOP setting remains unresolved by this theorem. It does not exclude interpretations used only for relative top termination, where strict monotonicity of every symbol is not required. In particular, it does not settle the top-removal approach in Yolcu–Aaronson–Heule's Lemma 3.18 [1]. Arctic interpretations, other strict orders, and higher dimensions are also outside its scope.

Real coefficients require a fixed positive decrement. For $\delta > 0$, put

$$x \succ_\delta y \iff x_0 \geq y_0 + \delta \text{ and } x_1 \geq y_1, \quad x, y \in \mathbb{R}_{\geq 0}^2.$$

Along a decrease, $\lfloor x_0/\delta \rfloor$ strictly decreases in $\mathbb{N}$, so $\succ_\delta$ is well-founded. Nonnegative affine maps preserve coordinatewise weak order. The bound $(M_s)_{00} \geq 1$ also preserves $\succ_\delta$, since

$$(F_s(x) - F_s(y))_0 = (M_s)_{00}(x_0 - y_0) + (M_s)_{01}(x_1 - y_1) \geq \delta.$$

A weak rule with first-offset gap $\varepsilon > 0$ decreases by at least ε at every nonnegative input. For a finite nonempty set of strict rules, the least such gap supplies a common $\delta > 0$. These observations explain the real-coefficient certificate format; ordinary strict inequality on the reals alone would not suffice. Theorem 1 prevents even this first choice of a strict rule. The preservation, evaluation, and well-foundedness claims above are formalized separately in `FullTwoSoundness.lean`.

3 Reduction of the digit matrices

Write A, B, C, D, E, F, G for the seven matrices and set $H = A + B + E + F + G$. The matrix parts of the six swaps give

$$AE \geq EA, \quad AF \geq EB, \quad AG \geq FA, \quad BE \geq FB, \quad BF \geq GA, \quad BG \geq GB. \quad (3)$$

Lemma 2. *Under the matrix hypotheses of Theorem 1, the five digit matrices are all upper triangular or all lower triangular in the original coordinates. Moreover, $A_{00} = B_{00} = E_{00} = F_{00} = G_{00} = 1$. This holds in both word orientations.*

Proof outline. Suppose $H_{01}, H_{10} > 0$. With $P = A + B$, the sum of the six nonnegative differences in (3) is $PH - HP$. The checked identities $\text{tr}(H^k(PH - HP)) = 0$, for $k = 0, 1$, force every entry of this commutator to vanish: $k = 0$ treats the diagonal, and $k = 1$ treats the off-diagonal entries. Each of the six differences therefore vanishes separately. The resulting exact swaps are incompatible with the boundary inequalities, as follows.

If $\det F = 0$, the exact relations yield a common positive left row. The rank-one identities and the checked two-dimensional stationary-vector lemma supply the corresponding right actions. Applying $BD \geq GD$ and $CF \geq CA^2$ gives $\alpha \geq \beta \geq \alpha^2$, with $\alpha \geq 1$. Thus both eigenvalues are one. Positivity forced by $H_{01}, H_{10} > 0$ contradicts the rank-one trace and the first-diagonal lower bound.

If $\det F \neq 0$, determinant identities give either $\det A = \det B = 0$ or nonsingularity of all five digits. For singular A, B , their traces coincide. When $A = B$, the common rank-one argument applies. Otherwise, writing this positive trace as α , exact swaps force

$$AB = \alpha A, \quad BA = \alpha B, \quad F = \beta I, \quad E = (\beta/\alpha)A, \quad G = (\beta/\alpha)B.$$

The boundaries give $\alpha^2 \leq \beta \leq \alpha$; hence $\alpha = \beta = 1$. The diagonal bounds and the two product identities then force all four mixed off-diagonal products to be zero, contradicting the assumed positivity of both off-diagonals of H .

In the nonsingular case $A = B$ forces $E = F = G$. The inequalities $AD \geq D$, $ED \leq AD$, $CE \geq CA^2$, together with $AE = EA$, give an entrywise contradiction to $H_{01}, H_{10} > 0$. If $A \neq B$, an exact classification gives $\mu, t > 0$ with

$$\begin{aligned} \text{tr } A = \text{tr } B = 3\mu, \quad \det A = \det B = 2\mu^2, \\ E = t(2A - \mu I), \quad F = t(A + B - \mu I), \quad G = t(2B - \mu I). \end{aligned}$$

Put $U = A - \mu I$, $V = B - \mu I$. These matrices are nonnegative and satisfy $U^2 = UV = \mu U$, $VU = V^2 = \mu V$. The boundary inequalities give $3t \leq 2$ and, for $u = (CU)_{00}$, $v = (CV)_{00}$,

$$u + v \leq 3tu, \quad 2\mu(u + v) \leq 3tv, \quad u + v > 0.$$

Consequently $\mu \leq \frac{1}{2}$. The trace and diagonal bounds force $\mu = \frac{1}{2}$ and $U_{11} = V_{11} = 0$, contradicting $(U + V)^2 = 2\mu(U + V)$ and positive off-diagonals.

Thus H has a zero off-diagonal entry. Nonnegativity makes that entry zero in every digit. Expanding the triangular swap and boundary inequalities then forces the five first diagonals to be one; both zero and positive secondary-diagonal cases are included in the formal proof. Transposition converts the reversed matrix rules to the forward ones. $\square$

4 The lower triangular affine case

After Lemma 2, lower triangular digit maps have the form

$$F_s(x, y) = (x + h_s, q_s x + z_s y + k_s), \quad h_s, q_s, z_s, k_s \geq 0. \quad (4)$$

Their first offsets add under composition. The six weak swap gaps sum to zero, so all six vanish and

$$h_a = h_b = r, \quad h_e = h_f = h_g = u, \quad r, u \geq 0. \quad (5)$$

Proposition 3. *For lower digit maps (4), every first-offset gap vanishes in either orientation, with arbitrary admissible boundary maps.*

Forward orientation. The dynamic rule $bd \rightarrow gd$ gives $r \geq u$. Write the first row of C as (c_0, c_1) , with $c_0 \geq 1$. If $c_1 = 0$, the cf boundary gives $u \geq 2r$, hence $r = u = 0$, and every boundary gap is zero.

Suppose $c_1 > 0$. The lower rows of the digit matrix products define scalar affine maps $y \mapsto z_s y + q_s$. The matrix inequalities induce all the forward scalar constraints, with outer boundary the identity and inner boundary the constant D_{10}/D_{00} . The checked scalar classification gives precisely two possibilities: every $q_s = 0$, or every $q_s = v > 0$ and every $z_s = 0$.

If every $q_s = 0$, apply the scalar classification to $y \mapsto z_s y + k_s$. Put

$$\Delta_e = k_e - k_b, \quad \Delta_f = k_f - (1 + z_a)k_a, \quad \Delta_g = k_g - k_a - z_a k_b.$$

The original boundary offset inequalities give $c_1 \Delta_e \geq c_0(r - u)$ and $c_1 \Delta_f, c_1 \Delta_g \geq c_0(2r - u)$. Since $r \geq u$, these supply the remaining scalar weak constraints. The scalar result makes all three Δ 's zero. Hence $u \geq 2r$, so $r = u = 0$ and all eleven first-offset gaps vanish.

In the other case, $q_s = v > 0$ and $z_s = 0$. The second-offset ag inequality and first-offset cf inequality yield

$$k_a - k_f \geq v(r - u), \quad k_f - k_a \geq vr + (c_0/c_1)(2r - u).$$

Their sum forces $r = u = 0$. The remaining weak inequalities make all five k_s equal, so the three boundary gaps vanish as well. $\square$

Reversed orientation. The first-offset swap equalities still give (5); now the inner boundary $fc \rightarrow aac$ gives $u \geq 2r$. If $D_{01} = 0$, the outer dynamic comparison gives $r \geq u$, and the result follows. For $D_{01} > 0$, the matrix constraints yield a reversed scalar system for $z_s y + q_s$. Its checked classification gives either all $q_s = 0$ or

$$z_a = 1, \quad z_b = z_f = z_g = 0, \quad q_a = 0, \quad q_b = q_f = q_g = \beta \geq 0.$$

In the zero case, the scalar reversed theorem applied to $z_s y + k_s$ gives $k_a = 0$ and $k_b = k_g$. The outer db inequality then forces $r = u = 0$. In the exceptional case, $gb \rightarrow bg$ gives $k_g \geq k_b$; combining this with $db \rightarrow dg$ again gives $r = u = 0$ and $k_b = k_g$. The remaining swaps force $k_a = 0$. The two dynamic gaps are therefore zero, and the other nine gaps already vanish. $\square$

5 Upper normalization and aggregate cancellation

Upper digit maps with unit first diagonals have the form

$$F_s(x, y) = (x + q_s y + h_s, z_s y + k_s), \quad q_s, z_s, h_s, k_s \geq 0. \quad (6)$$

In the forward orientation, replace the boundary maps by

$$\hat{C}(x, y) = (x + \kappa y, 0), \quad \hat{D}(x, y) = (x, \tau), \quad \kappa = C_{01}/C_{00}, \quad \tau = (v_d)_1. \quad (7)$$

The outer boundary matrix and offset comparisons in coordinate zero are divided by $C_{00} > 0$. The dynamic offset differences are unchanged, and their normalized matrix differences are zero. Thus all weak rules are preserved, and zero first-offset gaps in the normalized model imply zero gaps in the original model. This leaves 22 nonnegative parameters.

For a digit word, composition in (6) is described exactly by

$$\begin{aligned} q_{uv} &= q_v + q_u z_v, & z_{uv} &= z_u z_v, \\ h_{uv} &= h_u + h_v + q_u k_v, & k_{uv} &= k_u + z_u k_v. \end{aligned} \tag{8}$$

In particular, the products $q_u k_v$ remain in every offset comparison.

Lemma 4. *Put*

$$\begin{aligned} p &= z_a + z_b - 2, & Q &= q_a + q_b, & R &= k_a + k_b, \\ r &= z_e + z_f + z_g - 3, & P &= q_e + q_f + q_g, & K &= k_e + k_f + k_g. \end{aligned}$$

The sums of the six swap gaps in the q, k, h coordinates are

$$G_q = rQ - pP \geq 0, \quad G_k = pK - rR \geq 0, \quad G_h = QK - PR \geq 0. \tag{9}$$

If $p < 0$ or $r < 0$, and $Q + P, R + K > 0$, all six affine swaps are exact.

Proof. Expanding (8) gives (9) and

$$RG_q + QG_k = pG_h, \quad KG_q + PG_k = rG_h.$$

A negative coefficient on the right forces $G_h = 0$. Adding the resulting equalities gives $(R + K)G_q + (Q + P)G_k = 0$, hence $G_q = G_k = 0$. Every individual gap in these sums is nonnegative. The secondary-slope swap gaps also sum to zero, completing the assertion. $\square$

Two checked degenerate lemmas dispose of $Q + P = 0$ and $R + K = 0$ without any condition on the secondary slopes. In the second case all $k_s = 0$; the first offsets reduce to (5). If $\tau = 0$, bd and cf force $r = u = 0$. If $\tau > 0$, the matrix q -constraints form a reversed scalar system, which forces $q_a = 0$ and $q_b = q_g$; the same conclusion follows. The zero- q case follows from the forward scalar classification of the k -constraints.

The remaining proof therefore assumes positive total q and k . For exact swaps, a common form $q_s = \lambda(1 - z_s)$, $k_s = \eta(1 - z_s)$, with $\lambda, \eta > 0$, is impossible. Indeed, $H_s = h_s + \lambda k_s$ has binary value L and ternary value N . The dynamic rule gives $L \geq N$, while the cf matrix and offset comparisons give $N \geq 2L$. Thus $L = N = 0$, forcing every $k_s = 0$, a contradiction. These identities also treat boundary parameter values equal to zero.

6 Completion of the upper case and word reversal

Proposition 5. *Every normalized upper model satisfying all eleven weak forward rules has zero first-offset gap in every rule.*

Proof outline. First suppose $z_a, z_b > 0$. The exact slope comparisons make $z_e = z_f = z_g = s$. If $s > 0$, they also make $z_a = z_b = t$. For exact swaps between triangular matrices with diagonals (α, ρ) and (β, σ) , with $\alpha, \beta, \rho > 0$, the checked resonance calculation gives either equal binary off-diagonals and equal ternary off-diagonals, or $\rho = 2\alpha$, $\sigma = 3\beta$.

If $\min(s, t) < 1$, Lemma 4 makes the swaps exact. The q resonance $(t, s) = (2, 3)$ is unavailable, so $q_s = \lambda(1 - z_s)$ with $\lambda > 0$. The k calculation gives the common form from Section 5, except at $(t, s) = (1/2, 1/3)$. At that exceptional pair, the ce and cf matrix gaps force $\kappa = \lambda$; the projected

offset argument still gives $L \geq N \geq 2L$. If $s, t \geq 1$, the unit case is checked separately. For $1 \leq s \leq t$ with $t > 1$, the boundary inequalities give $P \geq 2Q$, whereas the aggregate inequality gives $P \leq \frac{3}{2}Q$, forcing every $q_s = 0$. For $s > t \geq 1$, the bg and bd secondary inequalities force $k_b = k_g = \tau = 0$, and the remaining swaps force every $k_s = 0$. If $s = 0$, exact swaps yield the same common form directly, even when the two positive binary slopes differ.

Now suppose at least one binary slope is zero. If the other exceeds one, the swap inequalities force every $q_s = 0$. Otherwise $p < 0$, so Lemma 4 again makes all swaps exact. When exactly one binary slope is zero, the common form follows directly. When both are zero, the same holds unless

$$(z_a, z_b, z_e, z_f, z_g) = (0, 0, 0, 1, 0), \quad q_a = q_b = q_e = q_g = q > 0, \quad q_f = 0.$$

In this exceptional branch, $k_e = k_a$, $k_f = 0$, $k_g = k_b$, and cf gives $\kappa \geq q$. Exact swaps and the bd, cf offset comparisons force $h_a + \kappa k_a \leq 0$, then $k_a = k_b = 0$, contradicting positive total k . This exhausts the slope cases; the degenerate lemmas give all eleven zero-gap equalities in every surviving model. $\square$

For reversed upper models, normalize instead

$$\widehat{D}(x, y) = (x + \lambda y, 0), \quad \widehat{C}(x, y) = (x, \sigma), \quad \lambda = D_{01}/D_{00}, \quad \sigma = (v_c)_1.$$

The same entrywise argument preserves all weak rules and transfers zero gaps back to the original boundaries. For an upper affine map, write its homogeneous matrix as

$$\mathcal{H}_s = \begin{pmatrix} 1 & q_s & h_s \\ 0 & z_s & k_s \\ 0 & 0 & 1 \end{pmatrix}.$$

If J reverses the three coordinates, $\Phi(\mathcal{H}) = J\mathcal{H}^\top J$ reverses products. It exchanges q_s, k_s and preserves z_s, h_s . Thus it preserves entrywise inequalities and the first-offset entry, while converting the normalized reversed rules into the forward rules. The boundary forms exchange; the symbol labels do not. This operation is used only after normalization, when every image is again affine.

Proof of Theorem 1. Apply Lemma 2. For lower digits use Proposition 3. For upper digits, normalize the boundaries and use Proposition 5; in the reversed orientation first apply Φ . The checked normalization lemmas transfer each of the eleven zero gaps back to the original interpretation. $\square$

7 Formal statement, verification, and scope

The formalization uses Lean 4 and mathlib [3, 4]. Matrices and offsets are functions on `Fin 2` with values in $\mathbb{R}$. The shared definition `Affine.comp X Y` is

$$(M_X M_Y, M_X v_Y + v_X),$$

so the formal composition convention is the one used throughout this paper. `Admissible` is exactly (2); `ForwardWeak` and `ReversedRealWeak` each contain eleven coefficientwise comparisons. The two zero-gap records each contain eleven equalities at offset index zero.

The main declarations in `FullTwoCoordinate.lean` are

Declaration in <code>CollatzResearch.FullTwo</code>	Conclusion
<code>forward_all_gaps_zero</code>	Eleven forward gaps vanish
<code>reversed_all_gaps_zero</code>	Eleven reversed gaps vanish
<code>full_two_coordinate_obstruction</code>	Both implications

Each declaration reports only `propext`, `Classical.choice`, and `Quot.sound`. No additional axiom or proof placeholder is used. The complete dependency closure, including the soundness lemmas below, contains 51 local modules. These were rebuilt in a fresh directory, and an additional audit queries the axioms of every public theorem and lemma, 225 declarations in total. The retained report records the exact source inventory, source and compiled-module hashes, compiler outputs, dependency revisions, and the final theorem audit. Existing local compiled files are excluded from that fresh build.

The fixed-gap soundness lemmas use the namespace `CollatzResearch.FullTwoSoundness`:

<code>admissible_preserves_gap</code>	Preservation under admissible maps
<code>weak_rule_gives_gap</code>	Evaluation of a weak rule with an offset gap
<code>gap_wellFounded</code>	Well-foundedness for a positive decrement

These three declarations are included in the combined fresh build and axiom audit. They do not formalize the prior Collatz equivalence or the general rule-removal theorem.

The checked environment is Lean 4.27.0 with mathlib revision

`a3a10db0e9d66acbebf76c5e6a135066525ac900`.

The main source has SHA-256

`429785555eecec277248380fe0c2e767c8c7303cf0e2712de29878dda2bd29a0`.

From the artifact root, the verification entry point is

```
python3 verify.py \
  --mathlib-root /path/to/pinned/mathlib \
  --build-root /path/to/temporary/builds \
  --report /path/to/fresh-check.json
```

The retained verification record is `verification/rebuild.json`. The claim-to-declaration map in `paper/lean-claims.md` identifies the supporting formal results. Appendix A exposes the actual definitions and theorem type, so the connection between the prose and the formal claim can be inspected directly.

The trust boundary consists of Lean’s kernel and the stated foundational axioms, together with the definitions shown in the appendix. Automated tactics construct proof terms that the kernel checks. The build rejects proof placeholders and additional axioms; it does not use numerical search results as proof. Kernel acceptance establishes the Lean proposition. The symbol table, appendix, and claim-to-declaration map address the separate question of whether that proposition is the one claimed in the paper. The Collatz equivalence is cited from prior work and is not asserted to have been re-formalized by this artifact.

7.1 Availability and reproducibility

The paper, Lean sources, verification script, retained compiler output, and claim-to-declaration map are available at

<https://github.com/x1xhlol/collatz-two-coordinate-obstruction>.

The repository records the source hashes and pins the dependencies. Reproduction uses a new output path to preserve the supplied audit record. The main obstruction has no coefficient cap, integrality assumption, or nonsingularity assumption. It does retain every admissibility condition in Section 2. In particular, the result neither proves the Collatz conjecture nor excludes the relaxed TOP setting.

Acknowledgments

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Use of AI

OpenAI Codex was used extensively for mathematical exploration, development of the Lean proofs and verification scripts, and preparation and revision of this manuscript. The proof claims are supported by the accompanying Lean artifacts, rather than by the system's assertions. This disclosure does not imply review or endorsement by the authors of the underlying rewriting system.

References

- [1] E. Yolcu, S. Aaronson, and M. J. H. Heule. *An Automated Approach to the Collatz Conjecture*. Journal of Automated Reasoning **67**(2), Article 15, 2023. <https://doi.org/10.1007/s10817-022-09658-8>.
- [2] E. Yolcu, S. Aaronson, and M. J. H. Heule. *Rewriting Collatz: implementation and rewriting systems*. <https://github.com/emreyolcu/rewriting-collatz>.
- [3] L. de Moura and S. Ullrich. *The Lean 4 Theorem Prover and Programming Language*. In Automated Deduction, CADE 28, pp. 625–635, 2021. <https://lean-lang.org/papers/lean4.pdf>.
- [4] The mathlib Community. *The Lean mathematical library*. In Certified Programs and Proofs, CPP 2020. <https://doi.org/10.1145/3372885.3373824>.

A Exact Lean declarations

The blocks below reproduce the checked source verbatim, with the original line numbers. The final block ends at the theorem's type; its proof term is omitted. Source paths are relative to the repository root.

The affine definitions and `ReversedRealWeak` are in `CollatzCertificate`. The two-coordinate definitions and final theorem are in `CollatzResearch.FullTwo`. Matrix notation is opened in both namespaces. The order on vectors is pointwise, so `Weak X Y` means that every coefficient of `Y` is at most the corresponding coefficient of `X`. Composition places the left argument outside the right argument.

Ambient types and the coefficient order. `formal/ReversedBinaryPowerClosure.lean`, lines 10–17.

```

10 variable {n : Type*} [Fintype n]
11
12 abbrev Vec (n : Type*) := n → ℝ
13 abbrev Mat (n : Type*) := Matrix n n ℝ
14
15 def EntrywiseLE (M N : Mat n) : Prop := ∀ i j, M i j ≤ N i j
16
17 local infix:50 " ≤ₘ " => EntrywiseLE

```

Affine composition and comparison. `formal/ReversedBinaryPowerClosure.lean`, lines 255–266.

```

255 structure Affine (n : Type*) where
256   matrix : Mat n
257   offset : Vec n
258
259 def Affine.comp (X Y : Affine n) : Affine n :=
260   (X.matrix * Y.matrix, X.matrix *ₘ Y.offset + X.offset)
261
262 def Affine.Nonnegative (X : Affine n) : Prop :=
263   (0 ≤ₘ X.matrix) ∧ 0 ≤ X.offset
264
265 def Affine.Weak (X Y : Affine n) : Prop :=
266   (Y.matrix ≤ₘ X.matrix) ∧ Y.offset ≤ X.offset

```

Two coordinates and admissibility. `formal/FullTwoBasic.lean`, lines 9–11.

```

9 abbrev Aff2 := Affine (Fin 2)
10
11 def Admissible (X : Aff2) : Prop := X.Nonnegative ∧ 1 ≤ X.matrix 0 0

```

Forward weak rules. `formal/FullTwoBasic.lean`, lines 21–32.

```

21 structure ForwardWeak (A B C D E F G : Aff2) : Prop where
22   ad : (A.comp D).Weak D
23   bd : (B.comp D).Weak (G.comp D)
24   ae : (A.comp E).Weak (E.comp A)
25   af : (A.comp F).Weak (E.comp B)
26   ag : (A.comp G).Weak (F.comp A)
27   be : (B.comp E).Weak (F.comp B)
28   bf : (B.comp F).Weak (G.comp A)
29   bg : (B.comp G).Weak (G.comp B)
30   ce : (C.comp E).Weak (C.comp B)
31   cf : (C.comp F).Weak (C.comp (A.comp A))
32   cg : (C.comp G).Weak (C.comp (A.comp B))

```

Forward offset equalities. `formal/FullTwoBasic.lean`, lines 34–45.

```

34 structure ForwardGapsZero (A B C D E F G : Aff2) : Prop where
35   ad : (A.comp D).offset 0 = D.offset 0
36   bd : (B.comp D).offset 0 = (G.comp D).offset 0
37   ae : (A.comp E).offset 0 = (E.comp A).offset 0
38   af : (A.comp F).offset 0 = (E.comp B).offset 0
39   ag : (A.comp G).offset 0 = (F.comp A).offset 0
40   be : (B.comp E).offset 0 = (F.comp B).offset 0
41   bf : (B.comp F).offset 0 = (G.comp A).offset 0
42   bg : (B.comp G).offset 0 = (G.comp B).offset 0
43   ce : (C.comp E).offset 0 = (C.comp B).offset 0
44   cf : (C.comp F).offset 0 = (C.comp (A.comp A)).offset 0
45   cg : (C.comp G).offset 0 = (C.comp (A.comp B)).offset 0

```

Reversed weak rules. Here the index type is finite and has decidable equality; the final theorem instantiates it with `Fin 2`. `formal/ReversedRealNormalization.lean`, lines 83–94.

```

83 structure ReversedRealWeak (A B C D E F G : Affine ι) : Prop where
84   da : (D.comp A).Weak D
85   db : (D.comp B).Weak (D.comp G)
86   ea : (E.comp A).Weak (A.comp E)
87   fa : (F.comp A).Weak (B.comp E)
88   ga : (G.comp A).Weak (A.comp F)
89   eb : (E.comp B).Weak (B.comp F)
90   fb : (F.comp B).Weak (A.comp G)
91   gb : (G.comp B).Weak (B.comp G)
92   ec : (E.comp C).Weak (B.comp C)
93   fc : (F.comp C).Weak (A.comp (A.comp C))
94   gc : (G.comp C).Weak (B.comp (A.comp C))

```

Reversed offset equalities. `formal/FullTwoBasic.lean`, lines 47–58.

```

47 structure ReversedGapsZero (A B C D E F G : Aff2) : Prop where
48   da : (D.comp A).offset 0 = D.offset 0
49   db : (D.comp B).offset 0 = (D.comp G).offset 0
50   ea : (E.comp A).offset 0 = (A.comp E).offset 0
51   fa : (F.comp A).offset 0 = (B.comp E).offset 0
52   ga : (G.comp A).offset 0 = (A.comp F).offset 0
53   eb : (E.comp B).offset 0 = (B.comp F).offset 0
54   fb : (F.comp B).offset 0 = (A.comp G).offset 0
55   gb : (G.comp B).offset 0 = (B.comp G).offset 0
56   ec : (E.comp C).offset 0 = (B.comp C).offset 0
57   fc : (F.comp C).offset 0 = (A.comp (A.comp C)).offset 0
58   gc : (G.comp C).offset 0 = (B.comp (A.comp C)).offset 0

```

The final theorem statement. `formal/FullTwoCoordinate.lean`, lines 30–34.

```

30 theorem full_two_coordinate_obstruction (A B C D E F G : Aff2)
31   (hA : Admissible A) (hB : Admissible B) (hC : Admissible C) (hD : Admissible D)
32   (hE : Admissible E) (hF : Admissible F) (hG : Admissible G) :
33   (ForwardWeak A B C D E F G → ForwardGapsZero A B C D E F G) ∧
34   (ReversedRealWeak A B C D E F G → ReversedGapsZero A B C D E F G)

```